NIKHIL PATIL

Baltimore, MD | 848-237-0128 | nikhilpatil1104@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Graduate Data Science student specializing in Predictive Modeling, Bayesian and Econometric analysis, and Data Analysis. Experienced in transforming large scale datasets into pricing and customer lifetime value insights using Python, SQL, and scikit-learn. Applied LLM tools for data analysis, experimentation & workflow automation. Skilled at communicating complex findings to both technical and non-technical stakeholders and support data-driven decision-making.

TECHNICAL SKILLS

- **Statistical & Modeling:** Bayesian Statistics, Econometrics, Predictive Modeling, Regression (OLS/Poisson/Multivariate), Hypothesis & A/B testing, Exploratory Data Analysis
- **ML & Data Tools:** Python, SQL, MongoDB (NoSQL), R (basic), AWS (basic), Pandas, NumPy, Scikit-learn, PyTorch, Git
- **BI & Visualization:** Power BI, Tableau, Interactive Dashboards, Business Intelligence, Model Deployment
- **Domain Knowledge:** Insurance Analytics, Pure Premium Modeling, Customer Lifetime Value, Pricing Strategy, Risk Assessment, ETL Pipelines
- **Core Competencies:** Resiliency, Insurance Knowledge, Modeling Programming Languages

PROFESSIONAL EXPERIENCE

Buildstone Realty Advisors | *Data Analyst* ***May 2025 - Jul 2025***

- Spearheaded development of Power BI dashboards visualizing CLV and risk metrics, reducing reporting time by 40% and accelerating cross-functional decision-making for marketing and product teams.
- Engineered resilient ETL pipelines for insurance analytics by integrating internal and external data sources, slashing manual handling time by 60% and increasing reporting workflow reliability.
- Collaborated with stakeholders to distill complex predictive modeling and Bayesian statistics concepts into actionable insights, elevating user experience and driving data-informed product improvements aligned with pricing strategy and underwriting guidelines.
- Directed analysis of large-scale datasets using Python and SQL, applying econometrics and Bayesian statistical techniques with scikit-learn to develop predictive models for customer lifetime value and pure premium modeling, improving decision-making efficiency by 35%.

Data Management Intern ***Sep 2024 - Apr 2025***

- Oversaw 50,000+ property records in MySQL and MongoDB, scripting validation routines that sustained 99% data accuracy and enforcing rigorous quality-control checks.
- Architected and implemented scalable ETL processes using Python and Git to transform raw internal and external data into structured formats for exploratory data analysis, enabling the analytics team to deliver insights 40% faster.
- Developed automated Python scripts for data cleaning, validation, and standardization, improving processing speed by 70% and ensuring robust datasets for downstream statistical modeling and hypothesis testing.

PROJECTS

Glaucoma Detection Model

- Engineered an end-to-to-end CNN-based glaucoma detection model from retinal fundus images that achieved 94% accuracy and improved sensitivity by 15% over the initial baseline.
- Leveraged transfer learning and an augmentation pipeline to increase effective dataset diversity 3×, reduce overfitting by 25%, and cut training time 40% with GPU optimization. Maintained version control and collaborative development workflows using Git.
- Conducted systematic hyperparameter tuning and evaluation using ROC-AUC, F1, recall metrics, boosting minority-class detection performance by 12%.

Climate Impact Prediction Using Regression Models

- Developed and calibrated multivariate OLS and Poisson regression models forecasting rainfall, sea-level rise, and extreme weather trends through 2050, improving prediction accuracy by 20% vs. historical baselines-methodology mirrors loss trend forecasting and insurance pricing model development.
- Constructed multi-decade environmental features (CO2, temperature anomalies, regional indicators), achieving $R^2 = 0.85$ (~30% gain over simpler models).
- Developed and deployed reusable machine learning pipelines using Python and Scikit-learn, improving country-level forecast precision by 22% and accele rating project runtime by 45%.

E-Commerce Price Comparison Platform

- Architected and deployed an end-to-end automated web scraping system using Selenium and BeautifulSoup, aggregating real-time pricing data across 10+ e-commerce platforms and processing 100,000+ product records-demonstrating large-scale data pipeline and ETL engineering capabilities.
- Designed and optimized a normalized MySQL relational database schema to store, index, and query 100,000+ product records, cutting average query response time by 40% through strategic indexing and query optimization.
- Developed intelligent price tracking and comparison algorithms that reduced customer search time by 50%, delivering actionable, data-driven purchasing recommendations through automated price trend analysis across competing platforms.

EDUCATION

University of Maryland Baltimore County **May 2027**

M.S., Data Science (GPA: 3.80)

- **Coursework:** Bayesian Statistics, Econometrics, Machine Learning & Gen AI , Big Data, Statistical Modeling

University of Mumbai **Jun 2025**

B.E., Electronics & Computer Science, Data Science (GPA: 3.33)

- **Coursework:** ML, Deep Learning, NLP, Database Management, AI

CERTIFICATIONS

- Generative AI Essentials: Using LLMs to Work with Data: by IBM SkillsBuild
- Enterprise Data Science in Practice: by IBM SkillsBuild
- Python For Beginners: by Udemy
- Automated Machine Learning For Beginners: by Udemy